Enterprise Network Setup - Project Documentation

1. IP Addressing Table

The following table shows the IP addresses, subnet masks, and default gateways assigned to the main network devices.

Device Name	Interface / Port	IP Address	Subnet Mask	Default Gateway
CCNA_LAB	Gig0/0.10 (VLAN 10)	192.168.10.1	255.255.255.0	N/A
	Gig0/0.20 (VLAN 20)	192.168.20.1	255.255.255.0	N/A
	Gig0/0.30 (VLAN 30)	172.16.1.1	255.255.255.0	N/A
	Gig0/0.50 (VLAN 50)	192.168.50.1	255.255.255.0	N/A
	Gig0/1 (WAN/Internet)	192.248.43.27	255.255.255.224	192.248.43.30
	Serial0/3/0 (DTE)	172.16.2.1	255.255.255.252	N/A
LAB_01	Gig0/0.100 (VLAN 100)	192.168.100.1	255.255.255.0	N/A
	Gig0/0.200 (VLAN 200)	192.168.200.1	255.255.255.0	N/A
	Gig0/0.500 (VLAN 500)	192.168.40.1	255.255.255.0	N/A

Device Name	Interface / Port	IP Address	Subnet Mask	Default Gateway
	Serial0/3/0 (DCE)	172.16.2.2	255.255.255.252	N/A
S2 (Switch)	VLAN 500	192.168.40.2	255.255.255.0	192.168.40.1
REMOTE_NET	Internet (WAN)	172.16.1.2	255.255.255.0	172.16.1.1
(Wireless Router)	LAN (Router IP)	192.168.30.1	255.255.255.0	N/A

2. VLAN Configuration

The table below lists the Virtual LANs (VLANs) created on both switches and their assigned switch ports.

Switch	VLAN ID	VLAN Name	Network Address	Allocated Ports
S1	10	STUDENTS	192.168.10.0 /24	Fa0/5 - Fa0/10
	20	STAFF	192.168.20.0 /24	Fa0/11 - Fa0/15
	30	REMOTE_NET	172.16.1.0 /24	Fa0/21
	50	MANAGEMENT	192.168.50.0 /24	-
	99	NATIVE	-	Gig0/1 (Trunk)
S2	100	STUDENTS	192.168.100.0 /24	Fa0/5 - Fa0/10
	200	STAFF	192.168.200.0 /24	Fa0/11 - Fa0/15

Switch	VLAN ID	VLAN Name	Network Address	Allocated Ports
	500	MANAGEMENT	192.168.40.0 /24	VLAN Interface
	990	NATIVE	-	Fa0/20 (Trunk)

3. Network Protocols & Services

The core services and routing methods running on the enterprise network:

- **DHCP:**
    - **CCNA_LAB Router:** Configured to automatically distribute IP addresses for VLAN 10 (STUDENTS_POOL) and VLAN 20 (STAFF_POOL).
    - **LAB_01 Router:** Configured to automatically distribute IP addresses for VLAN 100 (STUDENTS_POOL_2) and VLAN 200 (STAFF_POOL_2).
    - **REMOTE_NET Router:** Provides IP addresses for the wireless laptops from the 192.168.30.0 /24 network.
- **NAT:**
    - Configured **PAT** on the CCNA_LAB router. It uses Gig0/1 as the outside interface to the internet, translating private IPs from Access-list 1 to a public IP.
- **OSPF Routing:**
    - Enabled OSPF Process ID 1 in Area 0 on both routers to share routing information across the internal networks.
    - The default route to the internet (0.0.0.0) is distributed to all devices using the default-information originate command on the CCNA_LAB router.

4. Security Configurations

- **Device Passwords:** Console, VTY (Telnet/SSH), and Privileged EXEC (enable secret) passwords are set on all networking devices. All passwords are encrypted using the service password-encryption command.

- **Warning Banner:** A MOTD banner is added to display a warning message against unauthorized login attempts.
- **Port Security:** Configured on the access ports of the S1 Switch. It allows a maximum of 1 MAC address per port, uses MAC-address sticky to learn the connected devices, and applies the violation restrict action if an unknown device is plugged in.
- **Unused Ports:** To improve physical network security, all unused ports on the S2 Switch are administratively disabled (shutdown).